

can be enhanced with AI to improve test stability and reduce maintenance. AI plugins can automatically adapt to changes in the web application, making test scripts more resilient to UI changes.

Enhance Selenium with AI: Integrate AI-based plugins such as Applitools Eyes or Mabl into your Selenium setup.

Dynamic Element Location: Utilize the AI to recognize elements on a page, even when changes occur, by learning from the historical interaction data.

Self-healing Tests: Implement self-healing mechanisms that automatically update test scripts when an element's selector changes.

Visual Regression Testing: Use AI to compare screenshots taken during tests with baseline images to detect visual discrepancies.

Continuous Learning: Allow the AI to learn from each test run to reduce false positives and improve test accuracy over time.

2) Testim.io

Testim.io leverages AI to author, execute, and maintain automated tests with speed and precision. It provides a platform for testers to create reliable and resilient end-to-end tests.

Create Automated Tests: Use Testim.io's visual editor to create automated tests by interacting with your application; the platform learns and creates the tests.

AI Optimization: The AI observes the DOM (Document

Object Model) changes over multiple test runs to optimize element locators.

Parallel Execution: Run your tests in parallel across different browsers and environments to speed up the testing process.

Analyze Flakiness: Let Testim.io's AI analyze test runs to detect flaky tests and recommend improvements or updates to test cases.

Maintenance: Benefit from automatic test maintenance as Testim.io adjusts tests in response to UI changes, reducing the need for manual updates.

3) Applitools

Applitools uses Visual AI to automatically validate the user interface of web and mobile apps across different devices and browsers, ensuring that the app appears as intended to all users.

Integrate with Test Framework: Add Applitools SDK to your existing test framework to capture visual checkpoints.

Set Baselines: Define and manage visual baseline images that your application's UI is expected to match.

Automated Visual Tests: Execute tests, and Applitools will capture UI snapshots to compare against the baselines.

AI-Powered Analysis: The AI performs a smart visual analysis, identifying genuine visual defects while ignoring insignificant differences.

Review and Update: Review discrepancies flagged by the AI, determine if they are expected changes, and update baselines as needed.

